

Helpful Tools & When to Use Them

A plain-English guide to choosing the right tool for common business and data tasks

Practical Tools & Guides | Built for real-world reference, learning, and repeatable work.

The goal

Use the simplest tool that reliably solves the problem. More technical is not automatically better.

1. Database engine vs database client

Type	Examples	What it does
Database engine	MariaDB, MySQL, PostgreSQL, SQL Server, Oracle	Stores data, enforces rules, executes SQL.
Database client	HeidiSQL, DBeaver, SSMS, pgAdmin, MySQL Workbench	Connects people to databases and lets them browse/run SQL.
Spreadsheet	Excel, Google Sheets	Flexible analysis, formulas, quick models.
Transformation tool	Power Query	Repeatable shaping and cleanup of files and tables.
BI / visualization	Power BI, Tableau	Interactive dashboards, measures, visual analysis.

2. Excel vs SQL vs Power Query

Task	Usually best starting point
Quick calculation or small ad hoc analysis	Excel
Repeatable cleanup of files	Power Query
Join many related tables	SQL
Check for orphan records	SQL
Build a quick pivot/report for a business user	Excel
Validate rules across related tables	SQL
Interactive management dashboard	Power BI or Tableau

3. Common database platforms

Platform	Useful context
MariaDB / MySQL	Closely related SQL dialects; common in web and business applications.
PostgreSQL	Strong standards support, advanced data types, JSON, analytical features.

Platform	Useful context
Microsoft SQL Server	Uses T-SQL; common in Microsoft enterprise environments.
Oracle	Enterprise database platform using Oracle SQL and PL/SQL.
SQLite	Lightweight embedded database for local apps and prototypes.

4. Common database clients

Client	Typical strength
HeidiSQL	Lightweight Windows client; useful with MariaDB/MySQL and other supported systems.
DBeaver	Cross-platform multi-database client.
SSMS	Microsoft's primary SQL Server management environment.
pgAdmin	Common PostgreSQL administration client.
MySQL Workbench	MySQL-focused database design and administration.

5. Supporting tools that save time

Tool / category	Helpful for
CSV/text editor	Inspecting delimiters, line endings, header rows, raw values.
Excel	Spot-checking source data, formulas, quick comparisons.
Power Query	Repeatable cleanup, file combinations, reshaping.
Git/GitHub	Versioning SQL scripts, documentation, portfolio publishing.
Word / PDF	SOPs, guides, training materials, controlled forms.
PowerShell	Repeatable Windows/file tasks and basic system inspection.

6. Troubleshooting mindset

1. Identify the failing layer: source file, client, database engine, transformation, query, or report.
2. Read the exact error or warning.
3. Inspect the smallest example that reproduces the problem.

4. Validate assumptions such as data type, status label, path, delimiter, date format, and relationship.
5. Fix the cause rather than masking the symptom.
6. Rerun a validation check after the fix.

7. Versions matter

Menus, button names, and feature availability change by software version. Good guides explain the concept first and treat exact click paths as version-specific details.